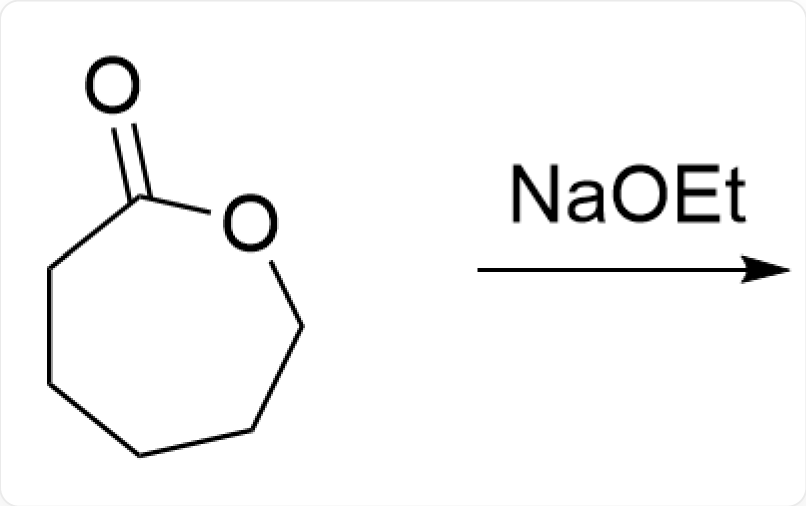
Question

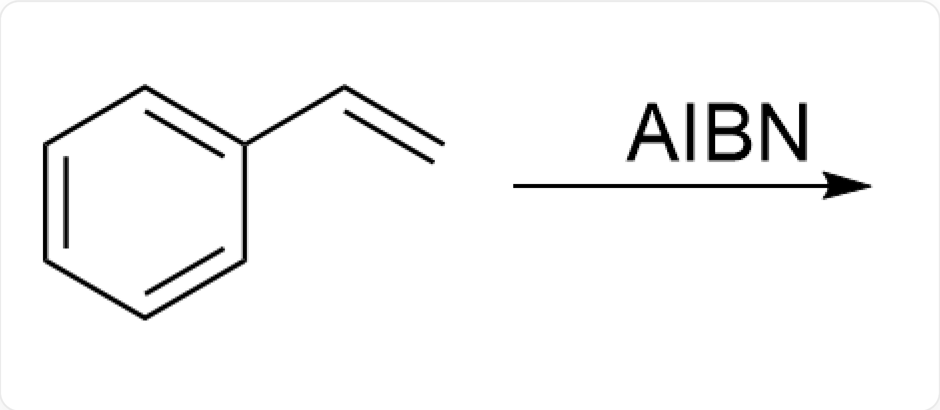
Here are several reactions. Which of them are living polymerizations?

Reaction 1:



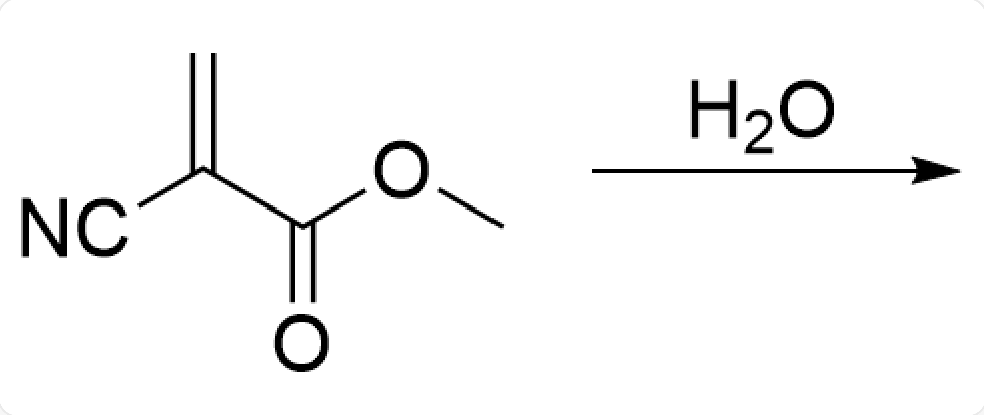
O=C1OCCCCC1>[Na]OCC>

Reaction 2:



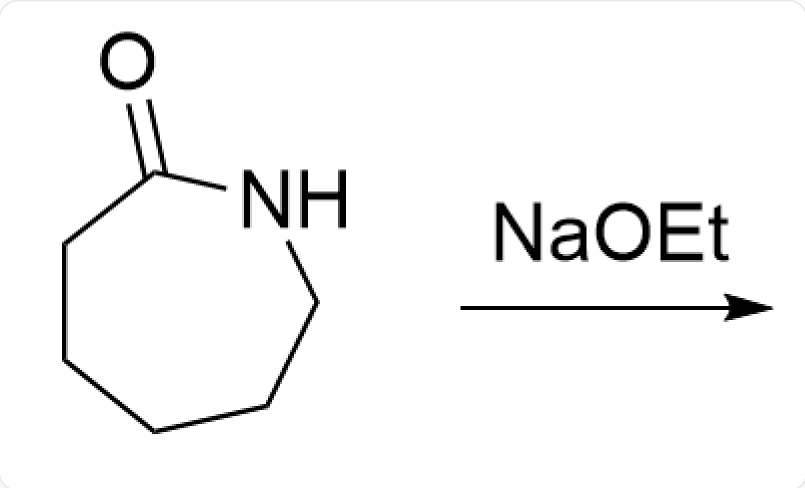
C=CC1=CC=CC=C1>N#CC(C)(C)/N=N/C(C)(C)C#N>

Reaction 3:



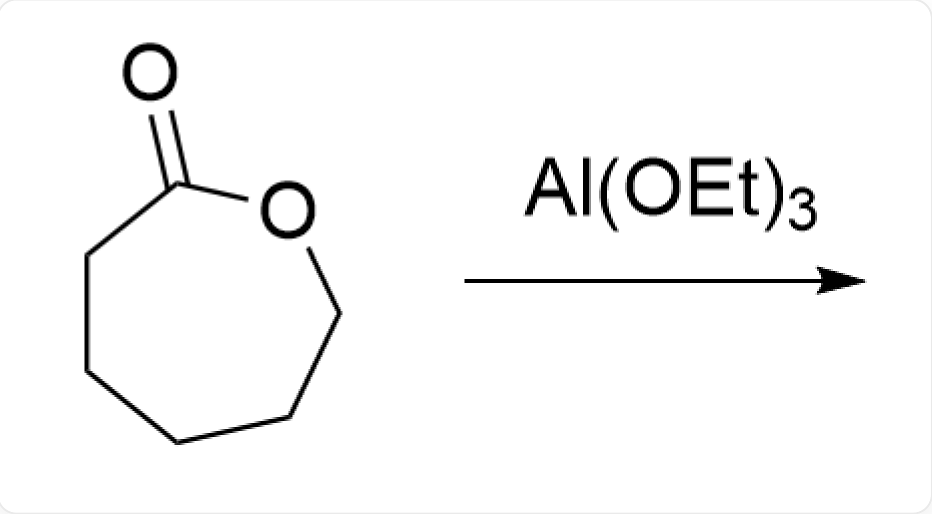
C=C(C#N)C(OC)=O>O>

Reaction 4:



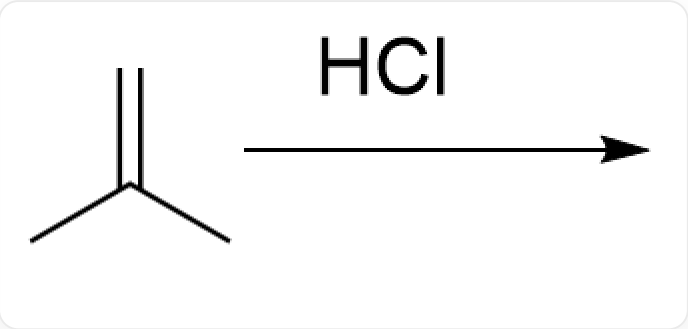
O=C1NCCCCC1>[Na]OCC>

Reaction 5:



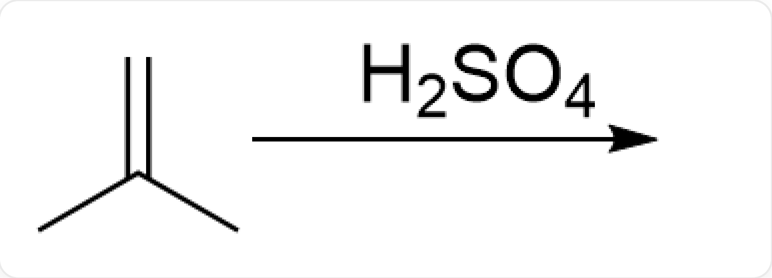
O=C1OCCCCC1>CCO[Al](OCC)OCC>

Reaction 6:



C=C(C)C>Cl>

Reaction 7:



C=C(C)C>O=S(O)(O)=O>

Reaction 8:

C=COC>II.I>

A. 1, 2, 3, 4, 5, 6, 7, 8

B. 1, 2, 3, 5, 6, 7, 8

C. 5,8

D. 7,8

E. 1, 2, 7, 8

F. 4,5,6,7,8

G. 1, 2, 3, 4, 5

H. 1, 2, 5, 6, 7, 8

I. No option indicated all the living polymerization reactions

Answer

Correct Answer: C

Detailed Explanation

In a living polymerization, the initiator generates a sufficient number of active centers, each of which reacts with monomers. The active centers are relatively stable with few side reactions until the reaction is terminated by the addition of a quenching agent. That is, living polymerization is characterized by fast initiation, slow propagation, no chain transfer, and no termination.

CHECKPOINT 1 PTS

Living polymerization features fast initiation, slow propagation, no chain transfer, and no termination

Reactions 1 and 4 involve chain transfer reactions where anions attack sites other than monomers, and thus are not living polymerizations.

CHECKPOINT 1 PTS

Reactions 1 and 4 involve chain transfer reactions

Reaction 2 is a radical polymerization with chain termination reactions, where the active chains are uncontrolled, and thus it is not a living polymerization.

CHECKPOINT 1 PTS

Reaction 2 has unstable active centers and involves chain termination reactions

In Reaction 3, water acts as a quenching agent for the active centers, leading to chain termination reactions, and thus it is not a living polymerization.

CHECKPOINT 1 PTS

Reaction 3 involves chain termination reactions

Reaction 5 is a coordination polymerization, where the O in the active center coordinates with Al, resulting in stable active centers and controllable polymerization, making it a living polymerization.

CHECKPOINT 1 PTS

Reaction 5 features O in the active center coordinating with Al, leading to stable active centers and controllable polymerization

Hydrochloric acid cannot initiate the polymerization of isobutene, so Reaction 6 is not a living polymerization.

CHECKPOINT 1 PTS

Hydrochloric acid cannot initiate the polymerization of isobutene

Reaction 7 is a cationic polymerization of isobutene, involving chain transfer reactions.

CHECKPOINT 1 PTS

Reaction 7 is a cationic polymerization of isobutene with chain transfer reactions

In Reaction 8, hydroiodic acid first iodinates the double bond to form an iodinated substrate. Iodine acts as a Lewis acid to activate the C-I bond, preventing the formation of free carbocations and thereby suppressing chain transfer and termination reactions in cationic polymerization, making it a living polymerization.

CHECKPOINT 1 PTS

Reaction 8 involves iodine as a Lewis acid activating the C-I bond, suppressing chain transfer and termination reactions